Unified Cloud-Based Framework for Hyperspectral and Multispectral Image Fusion Incorporating Nonlocal Principles and Tensor Decomposition

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***Abstract*— Hyperspectral image (HSI) super-resolution, which aims at improving the spatial quality of HSIs by fusing a low spatial resolution HSI (LR-HSI) with a high spatial resolution multispectral image (HR-MSI), has drawn significant attention. Numerous LR-HSI and HR-MSI (HSI–MSI) fusion algorithms have emerged in recent times, yet they suffer from a lack of generality and integration, which hampers their usability for nonexpert users. Moreover, these algorithms encounter significant challenges due to the exponential increase in remote sensing data volume. In this study, we propose a unified cloud-based framework for HSI–MSI fusion based on the general distributed alternating direction method of multipliers that incorporates nonlocal principles and tensor decomposition. The framework not only provides end-users with visualization modeling capabilities equipped with standard and comprehensive components but also enhances the parallel processing capabilities of cloud computing. We use a newly proposed nonlocal adaptive low-rank coupled tensor canonical polyadic (CP) decomposition algorithm as a case study to evaluate the performance of this framework. Specifically, we establish the LR-HSIs and HR-MSIs' relationship using order-4 coupled tensor CP decomposition and suggest an adaptive CP rank estimation method for achieving better super-resolution results. The experimental results on publicly available datasets demonstrate that the proposed parallel distributed optimization algorithm can achieve significant speedup with guaranteed accuracy. The proposed framework enables convenient and efficient processing of large-scale remote sensing data, effectively addressing the challenges associated with handling large data volumes. The source code of our method is released and available online at https://github.com/ZpWaitingForSunshine/DNAC4TCP/.**

***Index Terms*— Adaptive rank estimation, canonical polyadic (CP) decomposition, cloud-based distributed framework, hyperspectral and multispectral image fusion, nonlocal patch tensor (NPT).**

NOMENCLATURE

$\mathcal{X}$	N-order tensor $\mathcal{X} \in \mathbb{R}^{I_1, I_2, \ldots, I_N}$.
$\mathbf{X}, \mathbf{X}_k$	Matrix $\mathbf{X}, \mathbf{X}_k \in \mathbb{R}^{I_1 \times I_2}$.
$\mathbf{X}_n$	nth unfolding matrix for the tensor $\mathcal{X}$.
$\mathbf{x}, x$	Vector, scalar.
$\odot$	Khatri–Rao product.
$\times_{(n)}$	Mode-n tensor–matrix product.
$\mathrm{Tr}(\cdot)$	Matrix trace.
$< \mathcal{X}, \mathcal{Y} >$	Inner product of two tensors $\mathcal{X}$ and $\mathcal{Y}$.
$\|\mathcal{X}\|_F$	$\|\mathcal{X}\|_F = \sqrt{\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2}, \ldots, \sum_{i_N=1}^{I_N} x_{i_1 i_2, \ldots, i_N}^2}$.
$[\![\cdot]\!]$	Any tensor decomposition methods.

I. INTRODUCTION

HYPERSPECTRAL imaging plays a vital important role in remote sensing and computer vision applications, benefiting from the fact that the abundant spectral information of hyperspectral images (HSIs) enables a precise identification of the materials present in the scene [1], [2], [3]. However, there is always a tradeoff between the spatial and spectral resolutions due to the limitations of spectral imaging systems. For example, HSIs in general have high spectral resolution but low spatial resolution (LR). There have been considerable efforts devoted to enhance the spatial resolution of HSIs. Meanwhile, multispectral images (MSIs) usually have much higher spatial resolution but lower spectral resolution. Therefore, a popular strategy is to fuse an LR HSI (LR-HSI) with a high spatial resolution MSI (HR-MSI) to obtain a high spatial resolution HSI (HR-HSI). The LR-HSI and HR-MSI (HSI–MSI) fusion is widely used in various domains, including classification [4], [5], [6], [7], [8], [9], anomaly

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detection [10], [11], change detection [12], [13], and other remote sensing applications [14].

Previous studies [15], [16] typically required the flattening of the natural 3-D HSI data cube into a 2-D spectral matrix. However, this practice inevitably destroys the inherent structural correlation present within the HSI data. Therefore, many recent studies focus on tensor-representation-based approaches, which can better capture the multimodal characteristics and higher order correlations of data for HSI–MSI fusion [17]. For example, Li et al. [18] proposed an HSI–MSI fusion method based on low-rank tensor approximation and sparse representation. Moreover, nonlocal methods [19], [20] offer substantial advantages in HSI–MSI fusion by leveraging spatial self-similarity in the linear transformation domain. This approach effectively reduces artifacts and noise, enhances restoration quality, and ensures compatibility with multiple algorithms. Xu et al. [16] proposed a nonlocal patch tensor (NPT) sparse representation which extracts similar nonlocal patches to form an NPT and models NPTs in tensor–tensor (t-product)-based tensor sparse representation. However, different tensor-representation-based methods have variations in computational performance and fusion effectiveness. To the best of our understanding, there is currently a lack of automated solutions to customize and implement HSI–MSI fusion methods for users to select an appropriate decomposition method according to their specific requirements.

Meanwhile, with the significant development of Earth observation instruments, the amount of remote sensing data with higher spectral, spatial, and temporal resolution is increasing at an extremely high velocity. For instance, the Earth Observing System Data and Information System (EOSDIS) archived about 5.2 PB of data during 2018 and 2020 [21]. Thus, the tasks related to HSI data processing suffer from the challenge of heavy computation overheads and large-scale data storage requirements. In particular, since tensor decomposition requires more complicated processing than matrices' [22], the introduction of tensors into fusion applications undoubtedly leads to significant reduction in computational complexity.

Fortunately, cloud computing has attracted widespread attraction and became a promising solution to the efficient processing of remote sensing big data, mainly because of its outstanding capabilities in data storage and high-performance computing [21], [23], [24]. Qin et al. [25] proposed a cloud-based distributed parallel implementation of a remote sensing image fusion algorithm based on tensor canonical polyadic (CP) decomposition which achieves promising speedups and ensures HSI–MSI fusion quality. However, this approach requires individual parallel processing for each fusion algorithm, leading to increased encoding cost and limited flexibility. Wu et al. [26] proposed a scheduling-guided automatic processing method for massive classification of HSIs on cloud computing architectures. Although this method processes the tasks by the distributed platform Apache Spark, it is a coarse-grained approach that involves parallel processing solely at the algorithmic level. Recently, a new computing engine Ray [27] shines brightly in distributed computing, offering unique advantages in flexibility, superior performance, and ease of use when compared with Apache Spark.

In this study, we are committed to addressing the problem of efficient fusion of large-scale HSI data by developing a new cloud computing framework. This framework is designed for high-precision yet computationally costly nonlocal tensor decomposition algorithms. First, we establish a unified model for nonlocal low-rank tensor representation-based HSI–MSI fusion and offer users with standard and comprehensive components for visualization modeling. In addition, we build a distributed parallel alternating direction method of multipliers (ADMM) optimizer to solve the fusion problem by using the scalable computing engine Ray. Finally, we use a new nonlocal adaptive low-rank coupled tensor CP decomposition algorithm as a case study to evaluate the performance of our newly proposed HSI–MSI fusion framework. We incorporate the distributed two-stage K-nearest neighbors (KNN) algorithm, order-4 coupled CP tensor decomposition algorithm, and conjugate gradient (CG) algorithm in a modular manner. Specifically, the main contributions of this study can be summarized as follows.

1) We propose an HSI–MSI fusion paradigm based on the general distributed ADMM that incorporates nonlocal principles and tensor decomposition. The paradigm provides a set of standard and comprehensive components that can be flexibly combined and used according to requirements, offering users more choices and customization possibilities to meet different needs and scenarios.
2) We leverage cloud computing platforms for parallel processing in developing fusion components including tensor decomposition methods, NPT grouping, and so on. In this way, the HSI–MSI fusion paradigm is implemented in parallel and its efficiency is significantly improved.
3) We develop a distributed parallel nonlocal high-order coupled tensor CP decomposition as a case study. In particular, the relationship between the LR-HSI and the HR-MSI is established through order-4 coupled tensor CP decomposition which avoids the redundant unfolding process of NPTs while retaining spatial features. Furthermore, a novel tensor format Sparse Tensor is introduced to reduce the communication overhead and an adaptive CP rank estimation method to achieve better fusion results.

The experimental results on publicly available datasets demonstrate that the proposed parallel distributed optimization algorithm can achieve significant speedup with guaranteed accuracy. Moreover, the proposed framework enables convenient and efficient processing of large-scale remote sensing data, effectively addressing the challenges associated with handling large data volumes.

The remainder of this article is organized as follows. Section II presents the notations. Section III elaborates the newly proposed unified cloud-based framework for HSI–MSI fusion. In Section IV, a distributed parallel nonlocal order-4 coupled adaptive low-rank tensor CP decomposition is given as a case study. The experimental results are discussed in Section V. Finally, we draw a conclusion in Section VI.

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II. Notations and Problem Formulation

A. Introduction of Tensors

Following the notation convention of Sidiropoulos et al. [28], tensors are multidimensional structures with N modes denoted in Euler script letters, e.g., $\mathcal{X} \in \mathbb{R}^{I_1 I_2,\ldots,I_N}$, where I_N is the size of the mode-n dimension. For example, an order-3 tensor is a cube with three modes. A fiber is a vector obtained by fixing every tensor index but one. A slice is a matrix obtained by fixing every tensor index but two. For an order-3 tensor $\mathcal{X} \in \mathbb{R}^{i\times j\times k}$, $\mathcal{X}(:, j, k)$, $\mathcal{X}(i, :, k)$, and $\mathcal{X}(i, j, :)$ represent mode-1, mode-2, and mode-3 fibers separately. $\mathcal{X}(i, :, :)$, $\mathcal{X}(:, j, :)$, and $\mathcal{X}(:, :, k)$ represent the horizontal, lateral, and frontal slices, respectively. The mode-n unfolding of a tensor $\mathcal{X}$ is denoted as $\mathbf{X}_n \in \mathbb{R}^{I_n\times I_1 I_2,\ldots,I_{n-1}I_{n+1},\ldots,I_N}$ by rearranging the tensor mode-n fiber as the columns of the matrix. In addition, we have $\text{Unfold}_n(\mathcal{X}) = \mathbf{X}_n$ and $\text{Fold}_n(\mathbf{X}_n) = \mathcal{X}$. The mode-n tensor–matrix product is a projection operation of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 I_2,\ldots,I_n,\ldots,I_N}$ on a matrix $\mathbf{U} \in \mathbb{R}^{M\times I_n}$ to get a new tensor $\mathcal{Y} \in \mathbb{R}^{I_1 I_2,\ldots,M,\ldots,I_N}$ denoted as

$$\mathcal{Y} = \mathcal{X} \times_n \mathbf{U} \iff \mathbf{Y}_{(n)} = \mathbf{U}\mathbf{X}_{(n)}. \tag{1}$$

A more precise description of tensors and operations used in this article is given in nomenclature.

As a power tensor representation tool, CP decomposition [29] construct decomposes a tensor into a sum of rank-one components outer-products. For an order-4 tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3\times I_4}$, its CP decomposition construct is denoted as

$$\mathcal{X} = [\![\mathbf{U_1}, \mathbf{U_2}, \mathbf{U_3}, \mathbf{U_4}]\!] = \sum_{i}^{R} \mathbf{u}_{1,i} \circ \mathbf{u}_{2,i} \circ \mathbf{u}_{3,i} \circ \mathbf{u}_{4,i} \tag{2}$$

where $\mathbf{U_1} \in \mathbb{R}^{I_1\times R}$, $\mathbf{U_2} \in \mathbb{R}^{I_2\times R}$, $\mathbf{U_3} \in \mathbb{R}^{I_3\times R}$, and $\mathbf{U_4} \in \mathbb{R}^{I_4\times R}$. $\mathbf{U}_n$ is a factor matrix formed by stacking rank-one components $\mathbf{u}_{n,i}, i = 1,\ldots,R$ as columns into a matrix. R is the rank of the tensor $\mathcal{X}$ and refers to the minimal count of rank-one components. Equation (2) can also be written as

$$\begin{aligned}
\mathbf{X}_1 &= \mathbf{U_1}(\mathbf{U_4} \odot \mathbf{U_3} \odot \mathbf{U_2})^T \\
\mathbf{X}_2 &= \mathbf{U_2}(\mathbf{U_4} \odot \mathbf{U_3} \odot \mathbf{U_1})^T \\
\mathbf{X}_3 &= \mathbf{U_3}(\mathbf{U_4} \odot \mathbf{U_2} \odot \mathbf{U_1})^T \\
\mathbf{X}_4 &= \mathbf{U_4}(\mathbf{U_3} \odot \mathbf{U_2} \odot \mathbf{U_1})^T.
\end{aligned} \tag{3}$$

B. HSI–MSI Fusion Problem Formulation

In this article, the underlying HR-HSI is denoted as an order-3 tensor $\mathcal{X} \in \mathbb{R}^{W\times H\times S}$, where W, H, and S are the width, height, and spectral bands' numbers, respectively. Correspondingly, the LR-HSI is denoted as $\mathcal{Z} \in \mathbb{R}^{w\times h\times S}$ which is spatially downsampled w.r.t. $\mathcal{X}$ satisfying $W > w$ and $H > h$. The HR-MSI is denoted as $\mathcal{Y} \in \mathbb{R}^{W\times H\times s}$, which is spectrally downsampled w.r.t. $\mathcal{X}$ satisfying $S \gg s$. Mathematically, the LR-HSI and HR-MSI acquisition process [30] can be modeled as

$$\mathcal{Z} = \mathcal{X}\mathcal{S}\mathcal{H} + \mathcal{E}_h \tag{4}$$

and

$$\mathcal{Y} = \mathcal{R}\mathcal{X} + \mathcal{E}_m \tag{5}$$

respectively, where $\mathcal{S}$ denotes the spatial blurring operator, $\mathcal{H}$ denotes the spatial downsampling operator, and $\mathcal{R}$ denotes the spectral downsampling operator. $\mathcal{E}_h \in \mathbb{R}^{w\times h\times S}$ and $\mathcal{E}_m \in \mathbb{R}^{W\times H\times s}$ represent random modeling errors. The aim of the MSI-HSI fusion method is to computationally estimate $\mathcal{X}$ using $\mathcal{Y}$ and $\mathcal{Z}$. Its fundamental regularization model is given by

$$\min_{\mathcal{X}} \|\mathcal{Z} - \mathcal{X}\mathcal{S}\mathcal{H}\|_F^2 + \lambda\|\mathcal{Y} - \mathcal{R}\mathcal{X}\|_F^2 \tag{6}$$

where λ is a positive balancing parameter.

C. Unified Modeling Problem

Researchers have developed numerous fusion algorithms, but it can be challenging for users to directly use them due to issues such as inconsistent code execution environments and lack of standardized interfaces [21]. A low-code or even no-code framework [31] with unified fusion modeling is urgently needed to simplify the complexity of technology and address the inconsistency in computing environments. The framework should be carefully designed to prioritize compatibility between preceding and subsequent algorithms, enabling a seamless flow of data and results throughout the entire fusion process. In addition, the framework should provide a visual modeling approach that facilitates the selection of suitable algorithms to compose a fusion workflow that facilitates the user experience. Furthermore, it needs to be built on the cloud platform, which further increases the complexity.

D. Large-Scale HSI–MSI Fusion Problem

The challenge for large-scale HSI–MSI fusion is that the amount of remote sensing data with high spectral, spatial, and temporal resolutions is increasing at an extremely high speed with the significant development of Earth observation instruments [23]. Since tensor decomposition requires more complicated processing than matrices, its introduction in the fusion application undoubtedly increases the computational complexity [32]. In addition, for the NPT decomposition method, the patches are extracted in an overlapping manner, which further increases the data volume compared with the original LR-HSIs and HR-MSIs' [25]. Meanwhile, the smaller size of nonlocal patches' windows and steps lead to a larger NPT volume, which directly has an adverse effect on the fusion processing efficiency. Conventional single-machine computing platforms are no longer sufficient to handle such extensive processing demands of large-scale remote sensing datasets [33]. Cloud computing offers distinct benefits through its distributed storage infrastructure and the ability to scale computing resources [34], [35]. Consequently, cloud computing emerges as an optimal platform for the streamlined and effective processing of applications involving extensive remote sensing datasets. Although cloud computing addresses acceleration and storage issues, it introduces additional overheads in terms of communication during the fusion process [36].

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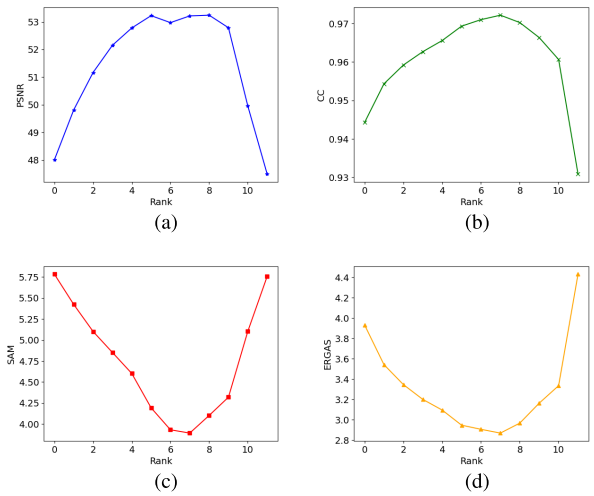


Fig. 1. Evaluation metrics of fusion results using different ranks with a step size difference of 10 from 10 to 120. (a) PSNR. (b) CC. (c) SAM. (d) ERGAS.

E. CP Rank Estimation Problem

Determining an optimal number of rank-1 tensors R, also known as tensor rank, is a critical challenge in CP tensor decomposition [37]. Unfortunately, there is no straightforward and practical approach to estimate R. After deploying our nonlocal-based fusion workflow, we selected 12 different ranks (from 10 to 120, with a step size difference of 10) to test the impact on fusion results. As shown in Fig. 1, the peak signal-to-noise ratio (PSNR), the relative dimensionless global error in synthesis (ERGAS), the cross correlation (CC), and the averaged spectral angle mapper (SAM) curves showing how the fusion results change with regard to the rank of the CP decomposition reveal that the fusion quality improves initially and then deteriorates as the rank increases, with the optimal rank being around 80. It is therefore apparent that too large or too small ranks can lead to poor image fusion performance.

III. Proposed Method

We develop a unified HSI–MSI fusion framework based on a general distributed ADMM, incorporating nonlocal principles and tensor decomposition. As shown in Fig. 2, a unified model for nonlocal low-rank tensor representation-based HSI–MSI fusion is established to establish a standardized modular workflow. The ADMM optimization is adopted for solving this model. Finally, a cloud-computing-based architecture is built to accelerate the HSI–MSI fusion process and offer users with standard and comprehensive components for visualization modeling.

A. Nonlocal Coupled Tensor Decomposition Model

According to the basic idea of nonlocal self-similarity, there may be multiple image patches with similar structure in an image, and these nonlocally self-similar image patches exhibit strong correlations. Generally, the spatial information (as a prior) guides the grouping of the underlying HR-HSI $\mathcal{X}$. Coincidentally, the HR-MSI $\mathcal{Y}$ (containing similar spatial information) can reflect the spatial distribution of $\mathcal{X}$. According to this idea, the HR-MSI $\mathcal{Y}$ is split into a set of overlapping 3-D patches $\{\mathcal{P}_i \in \mathbb{R}^{d_w \times d_h \times S}\}_{i=1}^{m}$, where $d_w \times d_h$ denotes the spatial shape of a patch and $m = (W - d_w + 1) \times (H - d_h + 1)$ denotes the number of patches. Then, similar patches can be grouped through a clustering method such as Kmeans [38] or Kmeans++ [39]. Each group has a size of G patches, except for the last group which may have a size smaller than G. After the grouping process, these 3-D patches are unfolded into a set of 2-D patches along the spectral mode and grouped into K groups according to similarity, or directly constructed as an order-4 tensor of size $d_w \times d_h \times G_k \times S$ in a group where G_k represents the kth NPT group.

Since the patches in a group are similar in terms of their spatial–spectral structures, $\mathcal{G}_k\mathcal{X}$ can be expressed through tensor decomposition as $[\![\mathcal{T}_k]\!]$, where $\mathcal{T}_k$ represents the factor. Thus, the HR-HSI $\mathcal{X}$ can be denoted as

$$\mathcal{X} = \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!] \tag{7}$$

which represents the reconstruction process of NPTs to form the HR-HSI.

Since the groups still keep the spectral information, the kth group of the HR-MSI $\mathcal{Y}$ can be correspondingly denoted as

$$\mathcal{G}_k\mathcal{Y} = \mathcal{G}_k\mathcal{X} \times_3 \mathbf{R} + \mathcal{G}_k\mathcal{E}_m \tag{8}$$

by referencing (5), where $\mathbf{R}$ is a spectral downsampling matrix. Thus, nonlocal principles and tensor decomposition can be used to estimate the target $\mathcal{X}$ by solving the following problem:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{T}_k} \|\mathcal{Z} - \mathcal{X}\mathcal{S}\mathcal{H}\|_F^2 + \lambda \sum_{k=1}^{K} \|\mathcal{G}_k\mathcal{Y} - [\![\mathcal{T}_k]\!] \times_3 \mathbf{R}\|_F^2 \\ &\text{s.t. } \mathcal{X} = \left(\sum_k \mathcal{G}_p^T \mathcal{G}_p\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!]. \end{aligned} \tag{9}$$

In the above model, the spectral structure is kept during the degeneration process from the HR-HSI to the HR-MSI in the nonlocal tensor representation. Meanwhile, the width, height, and nonlocal similarity features between the LR-HSI and the HR-MSI are transferred through the tensor decomposition.

B. ADMM Optimization

To solve the problem in (9), ADMM optimization [40] is adopted. Lagrange multipliers $\mathcal{M}$ are introduced and the augmented Lagrangian function in (9) can be written as

$$\begin{aligned} L_\mu(\mathcal{X}, \mathcal{T}_k) = &\|\mathcal{Z} - \mathcal{X}\mathcal{S}\mathcal{H}\|_F^2 + \lambda \sum_{k=1}^{K} \|\mathcal{G}_k\mathcal{Y} - [\![\mathcal{T}_k]\!] \times_3 \mathbf{R}\|_F^2 \\ &+ \frac{\mu}{2} \|\mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!]\|_F^2 \\ &+ \left\langle \mathcal{M}, \mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!] \right\rangle \end{aligned} \tag{10}$$

where μ is a positive penalty scalar. Specifically, the solution is given by the following steps.

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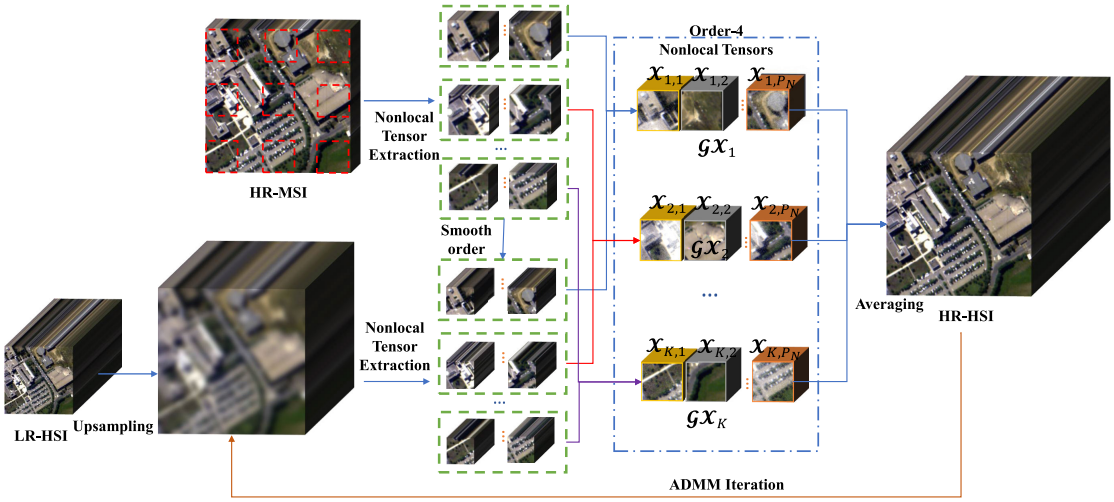


Fig. 2. Flowchart of the proposed HSI–MSI fusion incorporating nonlocal principles and tensor decomposition.

1) *Solving* $\mathcal{X}$*:* the subproblem L_μ with regard to $\mathcal{X}$ can be accomplished by

$$\min_{\mathcal{X}} \|\mathcal{Z} - \mathcal{X}\mathcal{S}\mathcal{H}\|_F^2 + \frac{\mu}{2}\|\mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!]\|_F^2 + \left\langle \mathcal{M}, \mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!] \right\rangle. \quad (11)$$

2) *Solving* $\mathcal{T}_k, k = 1, 2, \ldots, K$*:* the subproblem L_μ with regard to $\mathcal{T}$ can be accomplished by

$$\min_{\mathcal{T}_k} \lambda \sum_{k=1}^{K} \|\mathcal{G}_k \mathcal{Y} - [\![\mathcal{T}_k]\!] \times_3 \mathbf{R}\|_F^2 + \frac{\mu}{2}\|\mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!]\|_F^2 + \left\langle \mathcal{M}, \mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!] \right\rangle. \quad (12)$$

Due to the different tensor decomposition methods available, various approaches are required to solve $\mathcal{T}_k$. For instance, Xu et al. [41] model the fusion problem with CP tensor decomposition and adopt alternative least-squares (ALS) CP optimization to solve the fusion problem.

3) *Updating* $\mathcal{M}$ *and* μ*:* this can be done as follows:

$$\mathcal{M} = \mathcal{M} + \mu\left(\mathcal{X} - \left(\sum_k \mathcal{G}_k^T \mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathcal{T}_k]\!]\right) \quad (13)$$

μ is updated by

$$\mu = \min(\rho\mu, \mu_{\max}) \quad (14)$$

where $\rho \geq 1$.

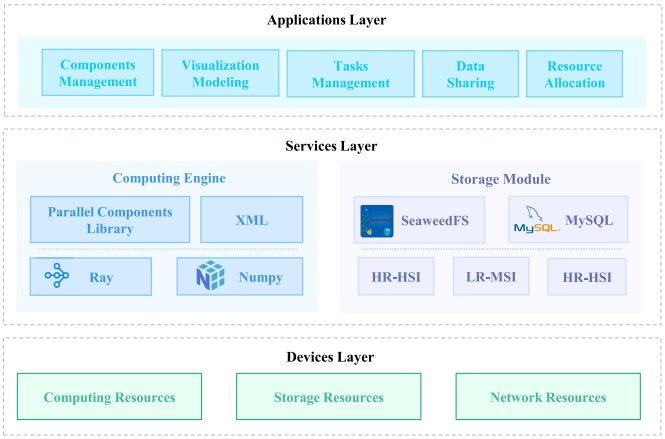


Fig. 3. Cloud -computing-based architecture for large-scale HSI–MSI fusion.

C. Unified Cloud-Based Fusion Framework

The ADMM solution of the nonlocal tensor decomposition model has high computational complexity due to the complexity of the model and the challenge of handling rapidly growing data. To address the above problem, we constructed a cloud-computing-based architecture for large-scale HSI-MSI fusion as shown in Fig. 3. Furthermore, we abstract these steps into a unified framework and provide users with a visualization modeling approach along with a parallel components library.

1) Devices Layer: The foundation of a cloud computing framework commences at the most elemental stratum, encompassing storage and server infrastructure. Specifically, the framework encompasses tangible assets such as servers, network apparatuses, and storage disks. Leveraging virtualization technology, these tangible resources coalesce into a reservoir of virtual resources for free allocation to achieve flexible deployment and dynamic adjustment of different applications and tasks, thereby improving system utilization and performance. The device layer serves as the bedrock for upper layers, furnishing robust and scalable resource environments that foster high availability for applications in the layers above.

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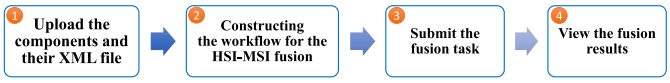


Fig. 4. Visualization modeling workflow of the entire fusion process.

2) Services Layer: The framework's services layer predominantly comprises modules dedicated to hyperspectral remote sensing big data storage and computational processes. For the storage, we use SeaweedFS [42] to store the LR-HSIs, HR-MSIs, and fused HR-HSIs in distributed clusters. SeaweedFS chooses to manage file volumes in the central master and lets volume servers manage files and metadata. SeaweedFS provides a simple and scalable way to store data and also is highly fault-tolerant, and designed to handle hardware failures gracefully. Besides, MySQL [43] is adopted to save the metadata of the original data.

Regarding the computing module, we adopt Ray as the distributed optimization tool, which has been widely used due to its big data processing capability and accessibility [27]. Ray is an excellent distributed computing framework and supports batch processing, real-time stream processing, etc. It consists of a core distributed runtime and a set of artificial intelligence libraries for simplifying machine learning computations.

3) Applications Layer: In the applications layer, we provide various optional parallel solutions for the NPT grouping operation and the subproblems related to solving $\mathcal{X}$ and $\mathcal{T}_k, k = 1, 2, \ldots, K$ by leveraging the capabilities of Ray for big data processing and the capacity of SeaweedFS for big data storage. Subsequently, we offer users a no-code visualization modeling approach for nonlocal HSI–MSI fusion. In detail, the components in the library are described using an Xtensible Markup Language (XML) file [44] for enabling a seamless flow of data and results throughout the entire fusion process. The XML file provides essential information for each algorithm, including its name, description, version, input parameters, output results, and other relevant details. With the availability of XML files, the visualization modeling approach for the fusion process shown in Fig. 4 can be implemented as follows.

1) *Upload the Components and Their XML Files:* The components in the fusion process are categorized into three groups: NPTs' grouping operation, results obtained after solving $\mathcal{X}$, and $\mathcal{T}_k, k = 1, 2, \ldots, K$. Each algorithm is optimized in a distributed parallel manner and described by an XML file. All these components collectively form a library that allows users to freely choose and combine them.
2) *Construct the Workflow for HSI–MSI Fusion:* We provide a visual interface with the components library for users to model their fusion workflow, allowing them to freely combine different components and match input and output parameters between preceding and subsequent nodes. A well-established workflow is reusable and does not require repetitive construction.
3) *Submit the Fusion Task:* Users can select data, fill in parameters, and submit tasks. We have also adapted the system to handle multiple tasks simultaneously, ensuring smooth execution of multiple tasks.
4) *View the Fusion Results:* The fusion results are stored in the SeaweedFS cluster. False color images of the obtained HR-HSIs are visualized in the system.

In this way, users can define their flowcharts based on their specific requirements. For example, users can make a choice between accuracy and speed based on their individual needs and preferences. In this regard, algorithms based on a fixed rank are typically faster than adaptive rank algorithms, but they may sacrifice some accuracy compared with the latter ones. In addition, they also benefit from the fast and efficient processing of large amounts of data.

IV. Case Study

To provide a clearer explanation of the unified cloud-based framework for HSI–MSI fusion, we use our newly proposed nonlocal adaptive coupled order-4 tensor CP (NAC4TCP) decomposition algorithm for HSI–MSI fusion as a case study. Specifically, we introduce high-order coupled CP tensor decomposition to construct the relationship between the LR-HSI and the HR-MSI. The augmented Lagrangian function for solving NAC4TCP can be detailed as

$$\begin{aligned} &L_\mu(\mathcal{X}, \mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k) \\ &= \|\mathbf{Z}_{(3)} - \mathbf{X}_{(3)}\mathbf{H}\mathbf{S}\|_F^2 + \lambda \sum_{k=1}^{K} \|\mathcal{G}_k\mathcal{Y} - [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{R}\mathbf{D}_k]\!]\|_F^2 \\ &\quad + \frac{\mu}{2}\|\mathcal{X} - \left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k]\!]\|_F^2 \\ &\quad + \left\langle \mathcal{M}, \mathcal{X} - \left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k]\!] \right\rangle. \end{aligned} \tag{15}$$

We incorporate the distributed two-stage KNN algorithm, order-4 coupled CP tensor decomposition algorithm, and CG algorithm [45] in a modular manner. In addition, we suggest a straightforward method to estimate the CP rank of each NPT. The corresponding workflow component console is shown in Fig. 5.

A. Distributed Two-Stage KNN Algorithm for NPTs Extraction

We adopt the KNN algorithm [46] as a simple yet effective NPTs' grouping method. Here, the overlapping patches greatly expand the data memory space consumption and network transmission overhead. Meanwhile, the KNN algorithm is computationally intensive especially when handling large-scale data. To solve this problem, we propose a distributed KNN algorithm including two stages: coarse grouping and fine grouping. In the coarse grouping stage, the overlapping patches are grouped into N groups where N is the number of parallel tasks. In the fine grouping stage, each group is processed by the KNN algorithm using individual worker nodes. In this way, the computational complexity and the amount of communication required for the subsequent operations (see Section IV-A) are reduced when compared with the naive algorithm.

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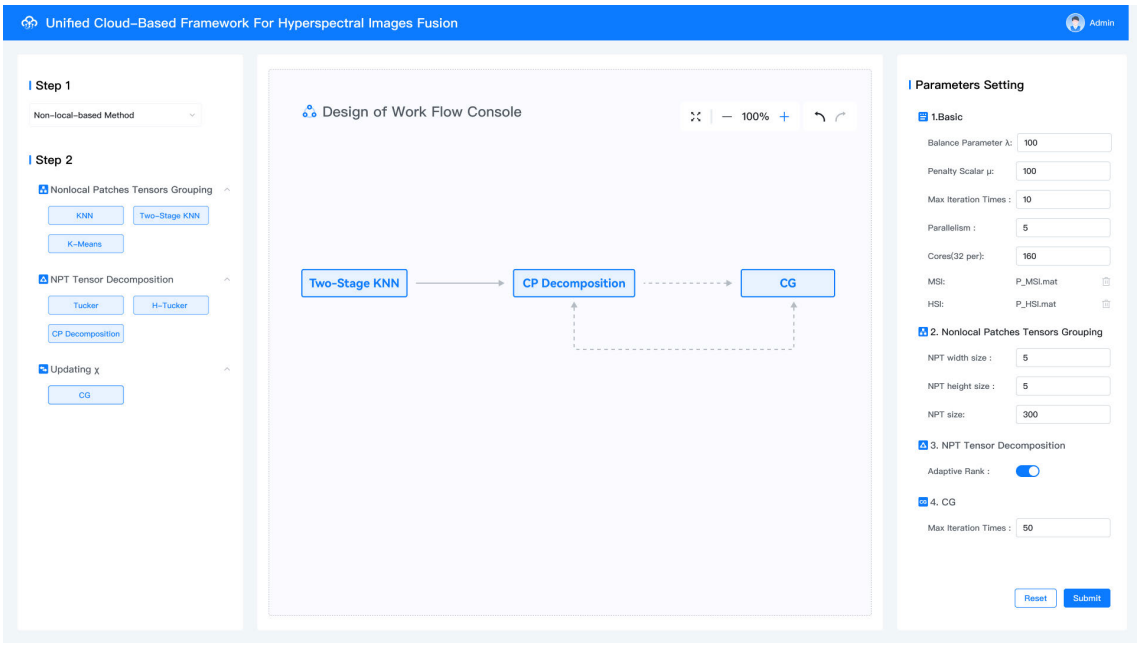


Fig. 5. Workflow component console including the distributed two-stage KNN algorithm, order-4 coupled CP tensor decomposition algorithm, and CG algorithms.

Algorithm 1 Two-Stage KNN Algorithm

Input: $\mathcal{Y}$, m, N, $rows$, $cols$, $patsize$, G
Output: The groups indices K

$Y_ref \leftarrow ray.put(\mathcal{Y})$
$indices \leftarrow array(2, rows \times cols)$
$indices[0] \leftarrow arange(rows \times cols)$
$indices[1] \leftarrow sys.maxsize$
$partitions_indices \leftarrow split_average(indices[0], G, num)$
$indices_set \leftarrow []$
for i in range$(N-1)$ **do**
  $indices_split_arrays \leftarrow array_split(indices, num, axis = 1)$
  $task_ids \leftarrow [knn.remote(item, rows, cols, patsize, indices[0][0], Y_ref)\ for\ item\ in\ indices_split_arrays]$
  $indices_list \leftarrow ray.get(task_ids)$
  $indices \leftarrow concatenate(indices_list, axis = 1)$
  $min_indices \leftarrow find_min_indices(indices, len(partitions_indices[i]))$
  $indices_set.append(indices[:, min_indices])$
  $indices \leftarrow delete(indices, min_indices, axis = 1)$
end for
$indices_set.append(indices)$
$task_ids \leftarrow [partitions_group.remote(item, rows, cols, patsize, Y_ref, G)\ for\ item\ in\ indices_set]$
$groups \leftarrow ray.get(task_ids)$

As shown in Algorithm 1, we share the LR-MSI $\mathcal{Y}$, which is cached in Ray's distributed shared-memory object store using ray.put(). We define a two-row matrix, where the first row records indices referring to the first pixel of each patch. The indices can be transformed through row $=\lfloor$indice[0][i]%rows$\rfloor$ and col $=\lfloor$(indice[0][i] − row)/rows$\rfloor$. The second row records the Euclidean distance from the test patch to the other patches. In the first stage, the indices are split averagely into N partitions by column. Meanwhile, the size of each partition (denoted as P) should be an integer multiple of G, except for the last partition. The formula for calculating P can be expressed as

$$P = \lceil \text{len(indices)}/G \rceil \times G.$$

The KNN.remote() tasks are performed remotely in a distributed manner, where each task is responsible for calculating the Euclidean distance from indices[0][0] to the other indices in each partition to get the indices and the Euclidean distance. It is worth mentioning that the patches in each task are extracted from the shared $\mathcal{Y}$ through the indices directly, which not only avoids the potential memory overflow during the extraction of the nonlocal tensors on the head worker but also reduces the network overhead from the head worker to the other workers. The initial memory burden $\mathcal{O}(md_wd_hs)$ in the head worker is effectively distributed across multiple nodes. The network transfer volume is reduced from $\mathcal{O}(md_wd_hs)$ to $\mathcal{O}(m + NWHs)$. After that, the P nearest patches' indices are found and saved in the indices_set. Then, those patches' indices are removed from the indices. In the next loop, we continue to find the P nearest patches' indices to indices[0][0] until the number of columns of indices is not larger than P. After the loop end, the remaining patches' indices in the indices_set are saved as the last NPT. The time complexity of the first stage is $\mathcal{O}((m + (N-2)P/2)(N-1)d_wd_hs)$.

In the next stage, partitions_group.remote() is performed in distributed manner and each task is responsible for performing more detailed grouping on the elements of its corresponding indices_set using the KNN algorithm. Benefiting from the data partitioning methods mentioned earlier, the assignment of tasks is balanced, ensuring an even distribution of workload

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Algorithm 2 Distributed Nonlocal CP Decomposition Factors Updating Algorithm

Input: LR-HSI $\mathcal{Z}$, HR-MSI $\mathcal{Y}$, $\mathbf{R}$, $\mathbf{H}$, $\mathbf{S}$, μ, d_w, d_h
Output: $\mathcal{X}$

1: $psActor \leftarrow ParameterServerActor.remote(nn)$
2: $factorActors \leftarrow [FactorActor.remote(item, Y_ref, patsize, rows, cols, nn)\ for\ item\ in\ K]$
3: $ray.get([actor.update.remote(X_ref, patsize, rows, cols, M_ref, Y_ref, lda, mu, R, shape)\ for\ actor\ in\ factorActors])$
4: $pieceHSI \leftarrow [actor.getEimg.remote()\ for\ actor\ in\ factorActors]$
5: $X \leftarrow ray.get(psActor.put_HR_HSI.remote(*pieceHSI))$

among the tasks. Finally, the grouped NPTs' indices are returned for the subsequent operations. The time complexity of the second stage is $\mathcal{O}((P/2+G)(P/G-1)d_w d_h s)$.

Taking (12) as a reference, the subproblem of updating the CP decomposition factors can be represented as

$$\begin{aligned}\min_{\Theta} \lambda \sum_{k=1}^{K} \|\mathcal{G}_k\mathcal{Y} - [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{R}\mathbf{D}_k]\!]\|_F^2 \\ + \frac{\mu}{2}\|\mathcal{X} - \left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k]\!]\|_F^2 \\ + \left\langle \mathcal{M}, \mathcal{X} - \left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1} \sum_k \mathcal{G}_k^T [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k]\!] \right\rangle\end{aligned} \tag{16}$$

where $\Theta = \mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k, k = 1, \ldots, K$. We adopt the ALS algorithm to minimize (16). Since the nonlocal patches are independent of each other, the subproblem in (16) can be converted into the following problem:

$$\begin{aligned}\sum_{k=1}^{K} \min_{\Theta} \lambda\|\mathcal{G}_k\mathcal{Y} - [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{R}\mathbf{D}_k]\!]\|_F^2 \\ + \frac{\mu}{2}\|\mathcal{G}_k\mathcal{X} - [\![\mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k, \mathbf{D}_k]\!] + \mathcal{G}_k\mathcal{M}/\mu\|_F^2.\end{aligned} \tag{17}$$

Therefore, the subproblem in (16) is naturally parallelizable to update the factors of each NPT. However, determining the optimal CP rank R for each tensor group is a major challenge. Underestimating R can lead to underfitting, while overestimating R can result in overfitting. To tackle this issue, we suggest a straightforward method to estimate the CP rank of each NPT. We drew inspiration from the approach proposed in the study [47] and gradually increased R until we achieved the optimal value by optimizing the following process to obtain a perfect fit. We achieve the optimal value by optimizing the process shown in Algorithm 2.

First, we create N FactorActors (its attributes and methods shown in Fig. 6) named factorActors to persist the updated factors of the NPTs and a ParameterServerActor (its attributes and methods shown in Fig. 7) named psActor to gather the results of functions Update in factorActors for the next iteration. In step 2, each factorActor receives the parameter patchList

FactorActor
patchList: List[Patch]
Init(group, patsize) : None **Update**(X_ref, patsize, rows,cols, M_ref, Y_ref, lda, mu, R, shape) : 3-D Sparse Tensor

Fig. 6. Actor class FactorActor: **patchList** represents the patches that each factor is supposed to deal with. The function Init initializes the factors and Update is performed for factors updating.

ParameterServerActor
HR_HSI: 3-D Tensor
Init(shape) : None **put_HR_HSI**(*HSI_Piece) : 3-D Sparse Tensor

Fig. 7. Actor class ParameterServerActor: **HR_HSI** is a 3-D tensor. The function Init initializes the HR-HSI as a zero 3-D tensor, and put_HR_HSI gathers and averages the HR-HSI pieces from all FactorActors.

Patch
indices : List[Integer] **A** : 2-D matrix **B** : 2-D matrix **C** : 2-D matrix **D** : 2-D matrix **curY** : 4-D tensor **curM** : 4-D tensor **Rank** : Integer = 1 default

Fig. 8. Class Patch: **indices** represents the patch's indices. **A**, **B**, **C**, and **D** represent the factors of CP decomposition. **curY** and **curM** represent the corresponding nonlocal patches of the Lagrange multipliers $\mathcal{M}$ and the HR-MSI. **Rank** is the initial rank of the CP decomposition.

(which represents a list of NPTs' indices) and initializes the corresponding class Patch (its attributes shown in Fig. 8) that it is responsible for performing updates on. Note that, thanks to the data partitioning methods in Algorithm 1, the tasks of NPTs' factors updating in factorActors are balanced.

As shown in Algorithm 3, during the updating process, each factorActor sequentially updates the factors according to the provided patchList. For each NPT, the iteration starts at rank $r = 1$. Following that, $\mathbf{A}_k^{(r)}, \mathbf{B}_k^{(r)}, \mathbf{C}_k^{(r)}, \mathbf{D}_k^{(r)}$ are updated sequentially. For each $\mathbf{A}_k$ subproblem, we have

$$\begin{aligned}\min_{\mathbf{A}_k} \lambda\|\mathcal{G}_k\mathbf{Y}_{(1)} - \mathbf{A}_k(\mathbf{R}\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k)^T\|_F^2 \\ + \frac{\mu}{2}\|\mathcal{G}_k\mathbf{X}_{(1)} - \mathbf{A}_k(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k)^T + \mathcal{G}_k\mathbf{M}_{(1)}/\mu\|_F^2.\end{aligned} \tag{18}$$

The solution to the problem in (18) can be calculated by (20), as shown at the bottom of the next page. The solutions to $\mathbf{B}_k$ and $\mathbf{C}_k$ are similar to that of $\mathbf{A}_k$'s and are shown in (21) and (22), at the bottom of the next page. For each $\mathbf{D}_k$ subproblem, we have

$$\begin{aligned}\min_{\mathbf{D}_k} \lambda \sum_{k=1}^{K} \|\mathcal{G}_k\mathbf{Y}_{(4)} - \mathbf{R}\mathbf{D}_k(\mathbf{C}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T\|_F^2 \\ + \frac{\mu}{2}\|\mathcal{G}_k\mathbf{X}_{(4)} - \mathbf{D}_k(\mathbf{C}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T + \mathcal{G}_k\mathbf{M}_{(4)}/\mu\|_F^2.\end{aligned} \tag{19}$$

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Algorithm 3 Function FactorActor.Update()

Input: LR-HSI $\mathcal{Z}$, HR-MSI $\mathcal{Y}$, $\mathbf{R}$, $\mathbf{H}$, $\mathbf{S}$, μ, d_w, d_h, $patchList$, $shape$
Output: HR-HSI $\mathcal{X}$

$\mathcal{X} \leftarrow zeros(shape)$
for k, $patch_k$ in $patchList$ **do**
  $r \leftarrow patch_k.Rank$
  $ind \leftarrow patch_k.Indices$
  $\mathcal{G}_k\mathcal{Y} \leftarrow$ indices2Patch(ind, Y_ref)
  $\mathcal{G}_k\mathcal{X} \leftarrow$ indices2Patch(ind, X_ref)
  $\mathcal{G}_k\mathcal{M} \leftarrow$ indices2Patch(ind, M_ref)
  Update $\mathbf{A}_k^{(r)}$ by (18)
  Update $\mathbf{B}_k^{(r)}$ by (21)
  Update $\mathbf{C}_k^{(r)}$ by (22)
  Update $\mathbf{D}_k^{(r)}$ by (19)
  $cube_k^r \leftarrow ktensor(\mathbf{A}_k^{(r)}, \mathbf{B}_k^{(r)}, \mathbf{C}_k^{(r)}, \mathbf{RD}_k^{(r)})$
  for i in $range(max(\mathcal{G}_k\mathcal{X}.shape))$ **do**
    Update $\mathbf{A}_k^{(r+1)}$ by (18)
    Update $\mathbf{B}_k^{(r+1)}$ by (21)
    Update $\mathbf{C}_k^{(r+1)}$ by (22)
    Update $\mathbf{D}_k^{(r+1)}$ by (19)
    $cube^{r+1} \leftarrow ktensor(\mathbf{A}_k^{(r+1)}, \mathbf{B}_k^{(r+1)}, \mathbf{C}_k^{(r+1)}, \mathbf{RD}_k^{(r+1)})$
    $variation \leftarrow norm(cube_k^r - \mathcal{G}_k\mathcal{Y}) - norm(cube_k^{r+1} - \mathcal{G}_k\mathcal{Y})$
    if $variation \leq 0.05 \times \sqrt{len(ind)}$ **then**
      $\mathbf{A}_k \leftarrow \mathbf{A}_k^{(r)}, \mathbf{B}_k \leftarrow \mathbf{B}_k^{(r)}, \mathbf{C}_k \leftarrow \mathbf{C}_k^{(r)},$
      $\mathbf{D}_k \leftarrow \mathbf{D}_k^{(r)}$; $break$;
    else
      $cube_k^r \leftarrow cube_k^{r+1}$; $r \leftarrow r+1$;
    end if
  end for
  $\mathcal{X} \leftarrow \mathcal{X} + cube_k^r$
end for
return $SparseTensor(\mathcal{X})$

The final solution can be obtained as shown in (23), at the bottom of the page. After that, we calculate the norm of the difference between the result cube$_k^r$ (also known as the target NPT) and $\mathcal{G}_k\mathcal{Y}$ and compare the norm of the difference with the precomputed norm by increasing rank $r = r+1$ denoted as variation. Based on experience, we stop the iterative process when the condition variation $\leq \sqrt{G}/20$ is satisfied.

Finally, each factorActor gathers the NPTs to form an HR-HSI which will be sent to psActor to reconstruct the underlying HR-HSI. However, the network overhead of transferring the HR-HSIs from all factorActors to psActor is significant. Fortunately, benefiting from the data partitioning methods in the first stage of Algorithm 1, the NPTs in each factorActor are spatially close. As a result, the merged HR-HSI in each factorActor often contains a large number of zero elements. Therefore, we introduce a new representation method for the gathered HR-HSI. As shown in Fig. 9, the SparseTensor format is a variant of the compressed sparse column (CSR) format [48], which uses a 2-D array: data and two 1-D arrays: column_indices and row_pointers. We compress the merged HR-HSI layer by layer according to the spectral dimension. The data for each spectral layer of the merged HR-HSI are stored in the respective layer of data. The column_indices and row_pointers only record the positions of nonzero elements of one spectral layer of the merged HR-HSI in CSR format, because each spectral layer is consistent in terms of zero and nonzero structures. This further reduces the network transmission. In summary, the network transmission can be reduced by approximately $\mathcal{O}((N-1)WHS)$ using the SparseTensor format.

SparseTensor
column_indices: List[Integer]
row_pointers : List[Integer]
data : Array[List[Double]]

Fig. 9. Actor class SparseTensor: **column_indices** and **row_pointers** are 1-D arrays that record the positions of nonzero elements of one spectral layer of the merged HR-HSI. **data** is a 2-D array that stores the data for each spectral layer of the merged HR-HSI.

B. Distributed Parallel CG Algorithm for Updating $\mathcal{X}$

To accelerate the process, the original problem can be divided into N subproblems by splitting the target $\mathcal{X}$ (along the spectral dimension) into N pieces, which are still a 3-D cube of width W and height H. The original problem can be converted into

$$\min_{\mathcal{X}^n, n=1,2,\ldots,N} \|\mathcal{Z}^n - \mathcal{X}^n\mathcal{S}\mathcal{H}\|_F^2 + \frac{\mu}{2}\|\mathcal{X}^n - \left(\left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1}\sum_k \mathcal{G}_k^T\mathcal{T}_k\right)^n\|_F^2 + \left\langle \mathcal{M}^n, \mathcal{X}^n - \left(\left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1}\sum_k \mathcal{G}_k^T\mathcal{T}_k\right)^n \right\rangle. \quad (24)$$

$$\mathbf{A}_k = (\mu\mathcal{G}_k\mathbf{X}_{(1)}(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k) + 2\lambda\mathcal{G}_k\mathbf{Y}_{(1)}(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{B}_k) + \mathcal{G}_k\mathbf{M}_{(1)}(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k)) \times (2\lambda(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{B}_k)^T(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{B}_k) + \mu(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k)^T(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{B}_k))^{-1} \quad (20)$$

$$\mathbf{B}_k = (\mu\mathcal{G}_k\mathbf{X}_{(2)}(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{A}_k) + 2\lambda\mathcal{G}_k\mathbf{Y}_{(1)}(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{A}_k) + \mathcal{G}_k\mathbf{M}_{(2)}(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{A}_k)) \times (2\lambda(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{A}_k)^T(\mathbf{RD}_k \odot \mathbf{C}_k \odot \mathbf{A}_k) + \mu(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{A}_k)^T(\mathbf{D}_k \odot \mathbf{C}_k \odot \mathbf{A}_k))^{-1} \quad (21)$$

$$\mathbf{C}_k = (\mu\mathcal{G}_k\mathbf{X}_{(3)}(\mathbf{D}_k \odot \mathbf{B}_k \odot \mathbf{A}_k) + 2\lambda\mathcal{G}_k\mathbf{Y}_{(1)}(\mathbf{RD}_k \odot \mathbf{B}_k \odot \mathbf{A}_k) + \mathcal{G}_k\mathbf{M}_{(3)}(\mathbf{D}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)) \times (2\lambda(\mathbf{RD}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T(\mathbf{RD}_k \odot \mathbf{B}_k \odot \mathbf{A}_k) + \mu(\mathbf{D}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T(\mathbf{D}_k \odot \mathbf{B}_k \odot \mathbf{A}_k))^{-1} \quad (22)$$

$$\mathbf{D}_k = (2\lambda\mathbf{R}^T\mathbf{R} + \mu\mathbf{I})^{-1}(2\lambda\mathbf{R}^T\mathcal{G}_k\mathbf{Y}_{(4)} + \mu\mathcal{G}_k\mathbf{X}_{(4)})(\mathbf{C}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T((\mathbf{C}_k \odot \mathbf{B}_k \odot \mathbf{A}_k)^T(\mathbf{C}_k \odot \mathbf{B}_k \odot \mathbf{A}_k))^{-1} \quad (23)$$

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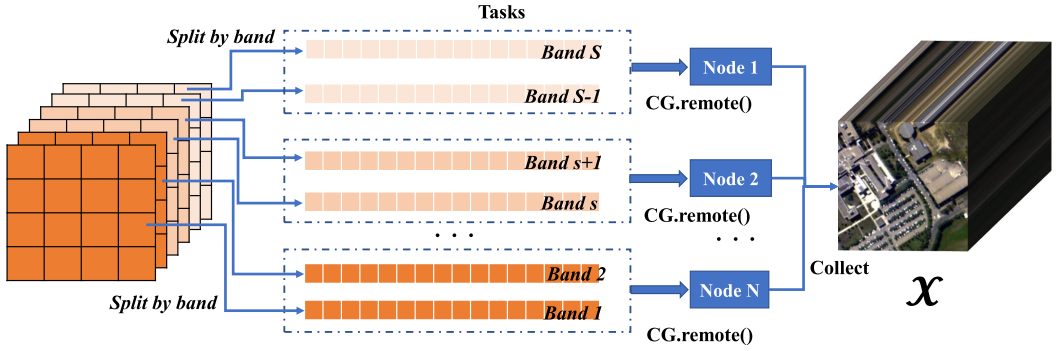


Fig. 10. Flowchart of the distributed parallel CG algorithm for updating $\mathcal{X}$.

Therefore, we can easily distribute the update of $\mathcal{X}$. Specifically, optimizing the subproblem with respect to $\mathcal{X}^n$ leads to the following linear system:

$$\mathcal{X}^n(2\mathcal{SH}(\mathcal{SH})^T + \mu\mathcal{I}) = 2\mathcal{Z}^n(\mathcal{SH})^T - \mathcal{M}^n + \mu\left(\left(\sum_k \mathcal{G}_k^T\mathcal{G}_k\right)^{-1}\sum_k \mathcal{G}_k^T\mathcal{T}_k\right)^n. \tag{25}$$

Here, we adopt the CG algorithm to solve the linear equation needed for updating $\mathcal{X}$. The flowchart of the distributed parallel CG algorithm for updating $\mathcal{X}$ is illustrated in Fig. 10.

V. Experiments

A. Platform and Datasets

To verify the framework's efficiency and practicality, we built a cluster with an application server, five storage servers, and five compute servers. The application server is equipped with an i7-9700K CPU at 3.60 GHz with eight cores, 32-GB memory, and 1-TB storage. Each compute server is equipped with an i9-9960X CPU at 3.10 GHz with 16 cores, 128 GB memory, and 1-TB storage. The software versions used include CentOS at 7.6, Python at 3.8, and Ray at 2.6. Physically, the storage servers and compute servers are shared.

Three widely used HSIs are used as benchmark images.

1) The first one is the University of Pavia dataset, captured over Pavia, Italy, by the Reflective Optics System Imaging Spectrometer (ROSIS) operated by the German Aerospace Agency. It contains 103 spectral bands after omitting the water vapor absorption bands. The size of the dataset is 610 × 340 × 103, resulting in a total volume of 162.98 MB.
2) The second one is the Washington DC Mall dataset, captured with the Hyperspectral Image Data Collection Experiment (HYDICE) sensor. It contains 210 bands covering the spectral range from 0.4 and 2.5 μm. The bands ranging between 0.9 and 1.4 μm are omitted from the dataset. The size of the dataset is 420 × 300 × 191, resulting in a total volume of 183.61 MB.
3) The third one is the Moffett Field dataset, captured over Moffett Field, California, by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). It contains 186 spectral bands after omitting the noisy bands. The size of this dataset is 1860 × 680 × 186, with a volume of 1.75 GB.

To simulate the real-world degradation as much as possible, the reference image is first blurred band by band with a Gaussian kernel to obtain the LR-HSI. Then, the blurred image is downsampled using a 9 × 9 downsampling kernel.

B. Detailed Implementation

In the experiments, our result is compared with CNMF [49], HySure [50], Sparse Bayes [51], CSTF [52], LTMR [53], and NCTCP [41]. Those methods are tested using their open-source codes and the default settings of parameters. To ensure the fairness of the experiment, the Gaussian kernel is used as the spatial degeneration operator in CSTF. In addition, the number of atoms in the spatial dictionaries is denoted as n_w and n_h. We set $n_w = 600$ and $n_h = 300$ for the University of Pavia dataset, $n_w = 400$ and $n_h = 300$ for the Washington DC Mall dataset, and $n_w = 1200$ and $n_h = 500$ for the Moffett Field dataset. Moreover, the size of the blurring kernel is set to 9 × 9 and its standard deviation is 2.12.

We choose the two-stage KNN, CP decomposition, and CG algorithm to form the fusion workflow NAC4TCP in the proposed platform. We set the initial parameters to $\mathcal{M} = \mathbf{0}$ and $\mu = 10^{-4}$. The number of iterations is set to 10. The window size $d_w \times d_h$ of each patch is 5 × 5, the step size is 1, and the size of the NPT is 300. The distributed version of NAC4TCP is developed in Ray and named DNAC4TCP. To verify the acceleration effect, we set the degree of parallelism and the number of nodes to be consistent, with a maximum degree of parallelism of 5. Furthermore, we consider the PSNR, ERGAS, SAM, and CC as the quality measures.

C. Fusion Results

1) University of Pavia Dataset: Fig. 11 shows the false color images (composed using bands 70, 30, and 5) obtained by different methods for the University of Pavia dataset. Specifically, Fig. 11(a) and (b) shows the false color images of the LR-HSI and the HR-MSI, respectively. Fig. 11(c) shows the false color image of the HR-HSI as reference. Fig. 11(d)–(k) shows the results of the comparison. Visually, the proposed NAC4TCP gives a more clear result (in spatial terms) than the other methods. Meanwhile, the distributed

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Fig. 11. Fusion results for the University of Pavia dataset. (a) LR-HSI. (b) HR-MSI. (c) Reference. (d) CNMF. (e) HySure. (f) SparseBayes. (g) CSTF. (h) LTMR. (i) NCTCP. (j) NAC4TCP. (k) DNAC4TCP.

TABLE I
QUALITY MEASUREMENTS FOR THE UNIVERSITY OF PAVIA DATASET

Method	PSNR	ERGAS	SAM	CC
CNMF	31.48	3.84	5.85	0.92
HySure	36.61	3.34	5.41	0.92
SparseBayes	37.80	3.17	5.93	0.93
CSTF	37.19	2.95	6.06	0.94
LTMR	37.31	2.87	4.95	0.94
NCTCP	37.15	2.36	4.38	**0.96**
NAC4TCP	39.60	**2.20**	**4.15**	**0.96**
DNAC4TCP	**39.71**	2.22	4.22	**0.96**

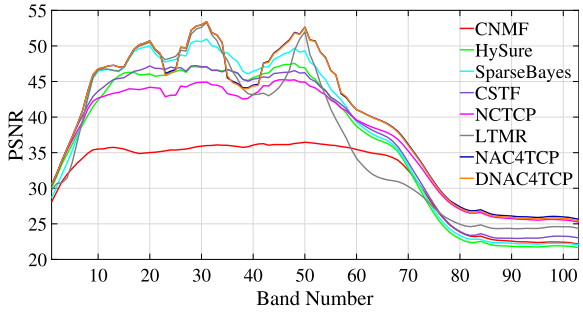


Fig. 12. PSNR comparison in each band of the University of Pavia dataset.

parallel version DNAC4TCP gets almost as good results as the serial version NAC4TCP.

Table I gives a more quantitative comparison. The proposed NAC4TCP and DNAC4TCP achieve better PSNR, ERGAS, SAM, and CC than the other methods. Table I demonstrates that our methods obtain a better preservation of both the spatial and spectral information. The proposed NAC4TCP outperforms CSTF in 2.41 dB in terms of PSNR which indicates that the similarity of nonlocal patches makes sense for HSI–MSI fusion. In addition, the proposed NAC4TCP and DNAC4TCP achieve better quality measures than NCTCP, which indicates that stacking the nonlocal similar patches into order-4 tensors and using the adaptive rank strategy can keep the inherent spectral information and is also effective for the preservation of spatial information. Referring to the PSNR comparison in all the bands in Fig. 12, the proposed method obtains the highest PSNRs in most bands indicating that the proposed method can achieve the best spectral and spatial preservation. The serial version NAC4TCP achieves similar performances as that of the distributed parallel version DNAC4TCP in all the quality measures. In theory, the two versions are equivalent in the solving process, differing only in the initial values.

2) Washington DC Mall Dataset: Fig. 13 shows the false color images (composed using bands 40, 30, and 5) obtained by different methods for the Washington DC Mall dataset. Specifically, Fig. 13(a) and (b) shows the false color images of the LR-HSI and the HR-MSI, respectively. Fig. 13(c) shows the false color image of the HR-HSI. Fig. 13(d)–(k) shows the results of the comparison. As in the University of Pavia dataset, the proposed NAC4TCP and DNAC4TCP give more clear spatial results than the other methods.

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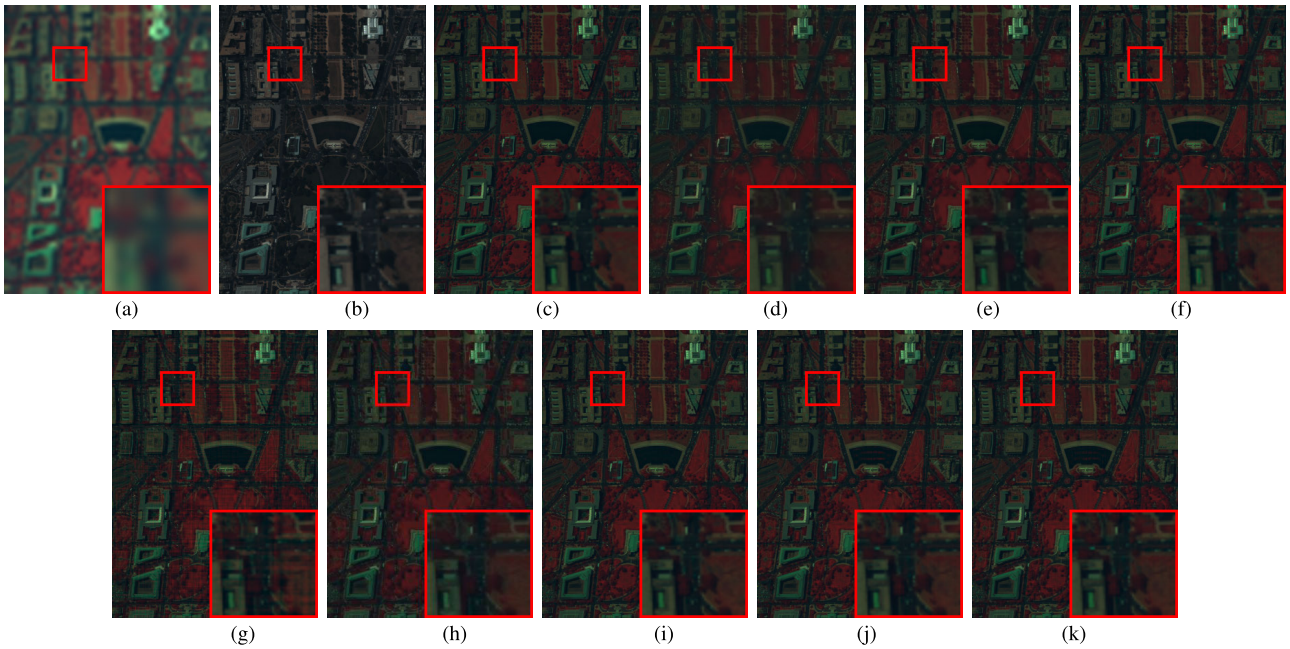


Fig. 13. Fusion results for the Washington DC Mall dataset. (a) LR-HSI. (b) MSI. (c) Reference. (d) CNMF. (e) HySure. (f) SparseBayes. (g) CSTF. (h) LTMR. (i) NCTCP. (j) NAC4TCP. (k) DNAC4TCP.

TABLE II
QUALITY MEASUREMENTS FOR THE WASHINGTON DC MALL DATASET

Method	PSNR	ERGAS	SAM	CC
CNMF	45.77	5.09	7.38	0.91
HySure	49.59	4.38	5.87	0.93
SparseBayes	51.43	3.56	4.47	0.95
CSTF	49.74	4.65	7.91	0.92
LTMR	52.14	3.84	6.16	0.95
NCTCP	53.03	**2.79**	3.88	**0.97**
NAC4TCP	**54.41**	2.82	**3.69**	**0.97**
DNAC4TCP	54.30	2.86	3.73	**0.97**

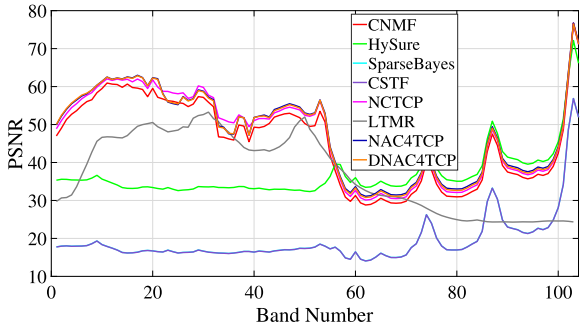


Fig. 14. PSNR comparison in each band of the Washington DC Mall dataset.

Table II provides a more quantitative comparison. Our proposed methods, NAC4TCP and DNAC4TCP, achieve superior PSNR, CC, and SAM compared with the other approaches. Regarding ERGAS, only a 0.03-unit improvement is achieved by NCTCP for the Washington DC Mall dataset. In Fig. 14, the PSNR comparison in each band of the Moffett Field dataset is shown. In the most bands, the proposed method achieves highest PSNRs, demonstrating its superiority. Similar to the results obtained with the previous dataset, our method excels in preserving both the spatial and spectral structures. Both the serial and parallel versions demonstrate exceptional performance.

3) Moffett Field Dataset: Fig. 15 shows the false color images (composed by using bands 70, 30, and 5) obtained by different methods for the Moffett Field dataset. Specifically, Fig. 15(a) and (b) shows the false color images of the LR-HSI and the HR-MSI, respectively. Fig. 13(c) shows the false color image of the HR-HSI. Fig. 13(d)–(k) shows the comparison results. The proposed NAC4TCP gives a more clear spatial result than the other methods.

Table III gives a more quantitative comparison. Our proposed methods, NAC4TCP and DNAC4TCP, achieve superior PSNR, ERGAS, CC, and SAM compared with the other approaches for the Moffett Field dataset. Note that the NCTCP method cannot run due to a memory overflow fault during the NPTs' grouping process, so "—" symbols are displayed in the table. However, our innovative approach has effectively addressed this issue by implementing a two-stage KNN method. In Fig. 16, the PSNR comparisons across each band are depicted, with the proposed method consistently attaining the highest PSNR values across the majority of bands. Similar to the results obtained with the previous datasets, our approach effectively maintains the integrity of both the spatial and spectral information. Both the serial and parallel versions demonstrate exceptional performance.

D. Influence of Parameters

In this section, we evaluate the influence of the adaptive rank strategy. We also develop a version without the adaptive rank strategy, named NC4TCP. We choose the rank ranging from 50 to 100 with step size 10 to perform NC4TCP. The size of the NPTs and the number of iterations are set to 300 and 10, respectively, for both NC4TCP and NAC4TCP.

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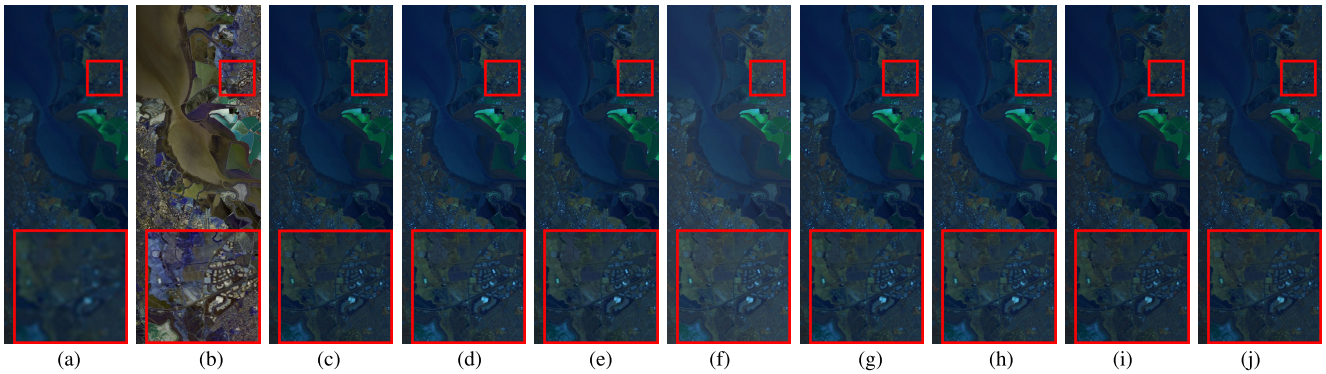


Fig. 15. Fusion results for the Moffett Field dataset. (a) LR-HSI. (b) MSI. (c) Reference. (d) CNMF. (e) HySure. (f) SparseBayes. (g) CSTF. (h) LTMR. (i) NAC4TCP. (j) DNAC4TCP.

TABLE III
QUALITY MEASUREMENTS FOR THE MOFFETT FIELD DATASET

Method	PSNR	ERGAS	SAM	CC
CNMF	44.72	4.00	2.74	0.97
HySure	44.37	4.08	3.73	0.97
SparseBayes	37.91	3.34	5.86	0.93
CSTF	42.75	4.37	5.46	0.92
LTMR	43.85	4.70	2.94	0.94
NCTCP	-	-	-	-
NAC4TCP	54.08	3.10	**2.27**	**0.97**
DNAC4TCP	**54.12**	**3.09**	**2.27**	**0.97**

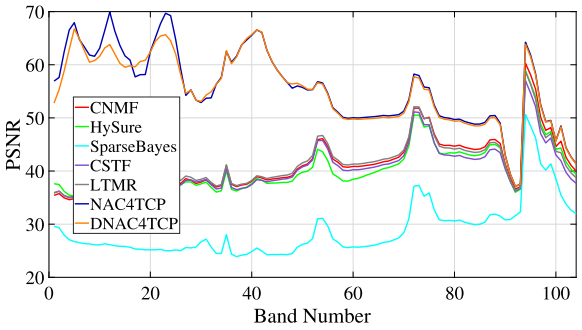


Fig. 16. PSNR comparison in each band of the Moffett Field dataset.

TABLE IV
IMPACT OF THE ADAPTIVE RANK STRATEGY ON THE ACCURACY OF RESULTS

Rank		50	60	70	80	90	100	Adaptive
University of Pavia	PSNR	38.67	39.24	39.60	39.76	39.72	38.95	**39.77**
	SAM	**4.09**	**4.09**	4.13	4.19	4.35	5.06	**4.09**
	ERGAS	2.25	2.24	2.25	2.26	2.30	2.64	**2.17**
	CC	0.96	0.96	0.96	0.96	0.96	0.96	**0.97**
	Time(s)	1377	1417	1434	1462	1498	1508	2793
Washington DC Mall	PSNR	52.85	53.44	53.94	54.19	54.09	53.48	**54.30**
	SAM	4.53	4.22	3.92	3.83	3.93	4.30	**3.72**
	ERGAS	3.20	3.06	2.94	2.89	2.95	3.14	**2.86**
	CC	0.96	**0.97**	**0.97**	**0.97**	**0.97**	**0.97**	**0.97**
	Time(s)	1358	1408	1415	1440	1473	1500	2347
Moffett Field	PSNR	51.35	51.90	43.28	19.41	23.39	27.45	**54.12**
	SAM	2.51	2.40	2.92	4.81	7.10	11.13	**2.27**
	ERGAS	3.23	3.21	9.62	129.18	72.75	51.90	**3.09**
	CC	**0.97**	**0.97**	0.90	0.16	0.28	0.36	**0.97**
	Time(s)	20836	21417	21260	22521	22187	23284	28410

TABLE V
IMPACT OF NPT SIZES ON THE ACCURACY OF RESULTS

NPT Size		200	300	400	500	600	Deviation
University of Pavia	PSNR	39.73	39.73	39.70	39.66	39.71	± 0.035
	SAM	4.16	4.17	4.22	4.27	4.22	± 0.055
	ERGAS	2.19	2.21	2.22	2.23	2.22	± 0.035
	CC	0.96	0.96	0.96	0.96	0.96	± 0.000
Washington DC Mall	PSNR	54.43	54.30	54.22	54.11	53.98	± 0.225
	SAM	3.62	3.73	3.74	3.79	3.78	± 0.085
	ERGAS	2.79	2.86	2.87	2.89	2.92	± 0.065
	CC	0.97	0.97	0.97	0.97	0.97	± 0.000
Moffett Field	PSNR	54.21	54.12	54.04	53.92	53.93	± 0.145
	SAM	2.25	2.27	2.28	2.29	2.30	± 0.025
	ERGAS	3.06	3.09	3.11	3.14	3.14	± 0.040
	CC	0.97	0.97	0.97	0.97	0.97	± 0.000

The results are compared with NAC4TCP in terms of PSNR, ERGAS, SAM, and CC, as shown in Table IV. With an increase in rank, the fusion performance initially improves, and then starts to deteriorate, with the best fusion performance observed at a rank of 80. Simultaneously, the execution time gradually increases with the rank. The adaptive rank strategy always achieves the best fusion performance across all three datasets, despite the fact that such strategy consumes the most time. In particular, the fixed rank method obtains relatively poor fusion results when dealing with very large images, as observed in the case of the Moffett Field dataset.

Furthermore, we conducted tests to evaluate the influence of varying NPT sizes on the results, aiming to validate the robustness of the adaptive rank strategy. In detail, we set the size of NPT ranging from 200 to 600 with a step of 100, respectively. The number of iterations is set to 10. By analyzing the deviation listed in the last column of Table V, we can observe that the changes in PSNR, SAM, ERGAS, and CC are small. Specifically, the PSNR and SAM values show little variation across different NPT sizes, indicating consistent performance. Among the considered datasets, the Washington DC Mall shows the largest variation in PSNR, but it is only by 0.45 dB. The University of Pavia dataset exhibits the smallest variation in PSNR, with only a 0.07-dB change. The differences in the SAM metric are even smaller, and the maximum change is only 0.17 in the Washington DC Mall dataset. The ERGAS metric also remains relatively stable, while the CC values remain consistent for each NPT size. Based on these findings, we can conclude that the NPT

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TABLE VI
IMPACT OF PARALLELISM ON THE ACCURACY OF RESULTS

Parallelism		1	2	3	4	5	Deviation
University of Pavia	PSNR	39.60	39.76	39.69	39.77	39.73	± 0.085
	SAM	4.15	4.21	4.21	4.09	4.17	± 0.060
	ERGAS	2.20	2.21	2.20	2.17	2.21	± 0.020
	CC	0.96	0.96	0.96	0.97	0.96	± 0.005
Washington DC Mall	PSNR	54.41	54.31	54.34	54.27	54.30	± 0.070
	SAM	3.69	3.69	3.68	3.68	3.73	± 0.025
	ERGAS	2.82	2.88	2.82	2.83	2.86	± 0.030
	CC	0.97	0.97	0.97	0.97	0.97	± 0.000
Moffett Field	PSNR	54.08	54.12	54.11	54.11	54.12	± 0.020
	SAM	2.27	2.27	2.27	2.27	2.27	± 0.000
	ERGAS	3.10	3.08	3.09	3.10	3.09	± 0.010
	CC	0.97	0.97	0.97	0.97	0.97	± 0.000

TABLE VII
EXECUTION TIMES (S) OF TWO-STAGE KNN, UPDATING $\mathcal{X}$, UPDATING FACTORS, AND TOTAL WORKFLOW

Nodes (Cores)		1(32)	2(64)	3(96)	4(128)	5(160)
University of Pavia	Two-Stage KNN	633.36	172.48	81.34	61.22	36.61
	Updating $\mathcal{X}$	3392.90	1632.06	1196.34	887.37	778.57
	Updating Factors	6988.94	3646.66	2591.22	2240.38	1963.03
	Total	11029.07	5465.00	3882.86	3202.97	2792.97
Washington DC Mall	Two-Stage KNN	235.02	63.51	33.02	21.92	16.42
	Updating $\mathcal{X}$	3629.05	2123.65	1622.09	1015.41	841.01
	Updating Factors	4919.08	2710.11	1909.71	1689.72	1473.99
	Total	8798.49	4912.65	3580.19	2742.46	2347.42
Moffett Field	Two-Stage KNN	24076.02	6575.45	3016.38	1734.33	1252.37
	Updating $\mathcal{X}$	40907.65	20557.00	13582.40	10551.70	8975.73
	Updating Factors	52940.16	35742.98	23847.89	20849.41	17986.52
	Total	118115.48	63066.91	40639.33	33327.60	28410.43

size has a negligible effect on the obtained results. The small changes in the evaluation metrics indicate the effectiveness of the adaptive rank strategy and that the chosen NPT sizes do not have a significant impact on the algorithm's performance.

E. Speedup and Comparisons

We tested the performance of DNAC4TCP on three datasets at different levels of parallelism using Ray. The detailed numerical data are presented in Table VI. As shown in the last column, the changes in the evaluation metrics are small and the parallel strategy has an insignificant impact on the algorithm's performance.

The execution times of the two-stage KNN, updating $\mathcal{X}$, and updating factors are recorded in Table VII. To visualize the acceleration effect more effectively, we plot these data graphically in Fig. 17. As illustrated in Fig. 17(a), the distributed parallel two-stage KNN algorithm achieves super-linear speedup with a maximum speedup ratio of 19× when the number of nodes is five and the dataset used is Moffett Field. Different from the naive KNN, the computational complexity of the distributed parallel two-stage KNN is indeed related to the number of nodes. As the number of nodes increases, the volume of data in each node decreases. For each query patch, the algorithm needs to calculate the distances between that patch and all other patches in each

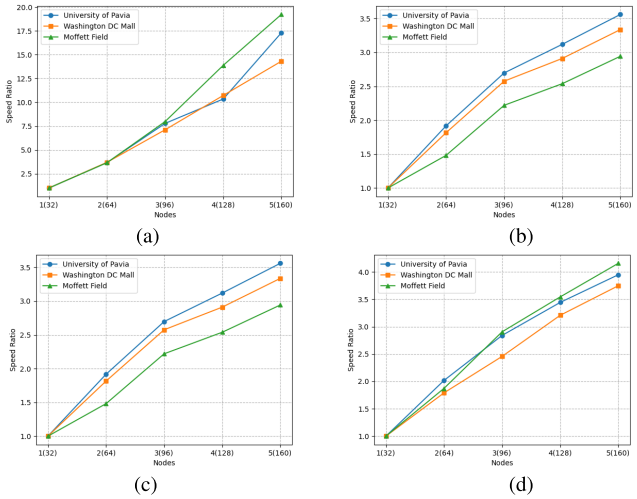


Fig. 17. Speedup ratios of (a) two-stage KNN, (b) updating $\mathcal{X}$, (c) updating factors, and (d) total workflow.

TABLE VIII
EXECUTION TIMES (S) OF NPTS GROUPING

Nodes (Cores)		1(32)	2(64)	3(96)	4(128)	5(160)
University of Pavia	Two-Stage KNN	633.36	172.48	81.34	61.22	36.61
	KNN	661.62	350.35	243.49	219.29	187.09
Washington DC Mall	Two-Stage KNN	235.02	63.51	33.02	21.92	16.42
	KNN	246.60	130.69	91.43	85.54	73.59
Moffett Field	Two-Stage KNN	24076.02	6575.45	3016.38	1734.33	1252.37
	KNN	25907.65	20557.00	8824.91	7277.87	6317.33

node. Thus, the complexity of each node decreases by a square factor as the number of nodes increases. The updating $\mathcal{X}$ and factors steps achieve a near-linear speedup, as shown in Fig. 17(b) and (c). As the number of nodes increases, the communication overhead experiences a sharp increase. This means that there is more data exchange and coordination required between the nodes, leading to increased communication time and potentially impacting the overall performance. Thus, the acceleration ratios start to slow down as the number of nodes increases. The total execution time is illustrated in Fig. 17(d). The three considered datasets all demonstrate excellent speedup and the acceleration trend is relatively stable, which benefits from our optimization in computing and communication.

Furthermore, to verify the effect of the two-stage KNN and the utilization of the SparseTensor format on acceleration and communication, we developed an alternative version without the two-stage approach and excluding the SparseTensor format. In Table VIII, we can observe the execution time of both the distributed parallel two-stage KNN and the naive KNN algorithms. It is evident that as the number of nodes increases, the execution time decreases gradually for both the algorithms. As shown in Fig. 18, compared with the naive KNN, the acceleration effect of the two-stage KNN algorithm becomes increasingly notable. This indicates that our designed two-stage KNN algorithm is effective.

In addition, the two-stage KNN algorithm brings an additional advantage in terms of communication during the collection of HR-HSI pieces among the nodes. During the

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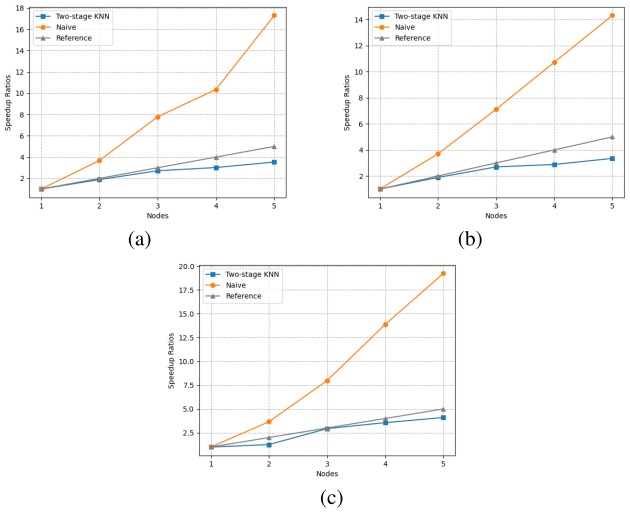


Fig. 18. Speedup ratios of the two-stage KNN method compared with the naive method. (a) University of Pavia. (b) Washington DC Mall. (c) Moffett Field.

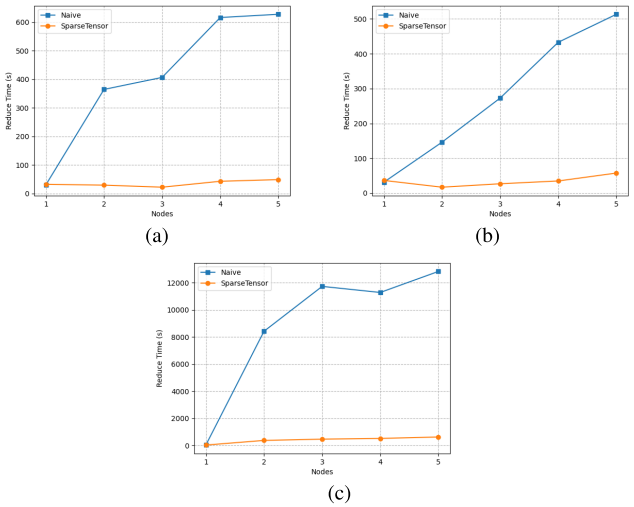


Fig. 19. Execution time of gathering and averaging pieces of the HR-HSI. (a) University of Pavia. (b) Washington DC Mall. (c) Moffett Field.

first stage of the two-stage KNN, the NPTs in each node are close in space, and there is a higher degree of overlap and also more areas with no data. The HR-HSI pieces in each node can be efficiently compressed by the SparseTensor format. The volumes of nonzero elements transferred without the SparseTenosr format will be an integer multiple of the observed HR-HSI's volume. But the SparseTensor format tensor stores only the nonzero elements and their corresponding indices, while ignoring the zero elements, effectively compressing the transmitted data volume, leading to significant savings in communication costs as shown in Fig. 19. As the degree of parallelism increases, the degree of overlap also increases and the average percent of zero elements in each node decreases, as shown in Fig. 20(a), while the volume of transferred nonzero elements increases in a sublinear rate, as shown in Fig. 20(b). As a result, the communication overhead is significantly reduced compared with the naive methods, as shown in Table IX.

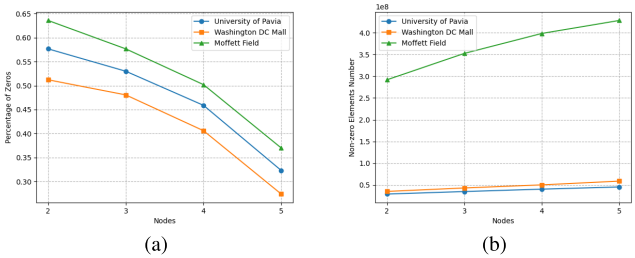


Fig. 20. (a) Percentage of zero elements for different degrees of parallelism. (b) Nonzero elements transferred for different degrees of parallelism.

TABLE IX
COLLECTION TIMES (S) OF THE HR-HSI AMONG THE NODES

Nodes (Cores)		1(32)	2(64)	3(96)	4(128)	5(160)
University of Pavia	Sparse Tensor	32.51	29.69	22.49	43.19	48.99
	Naive	30.69	364.50	406.65	616.17	627.46
Washington DC Mall	Sparse Tensor	36.48	17.10	26.95	34.96	57.59
	Naive	31.38	146.27	272.82	433.10	513.39
Moffett Field	Sparse Tensor	38.32	378.18	472.96	525.40	632.12
	Naive	39.14	8425.77	11733.84	11287.51	12837.06

VI. CONCLUSION

In this article, we propose a novel unified cloud-based framework for HSI–MSI fusion incorporating nonlocal principles and tensor decomposition. The framework provides an HSI–MSI fusion paradigm with visualization modeling offering users various optional components including but not limited to tensor decomposition methods. The ADMM optimization for the unified model is accelerated by leveraging the parallel processing capabilities of cloud computing. As a case study, we adopt a newly proposed nonlocal adaptive low-rank coupled high-order tensor CP decomposition algorithm for HSI–MSI fusion. It constructs the relationship between the LR-HSI and the HR-MSI through order-4 tensor CP decomposition. A straightforward method is introduced to estimate the CP rank of each nonlocal tensor for better fusion results. In conclusion, the framework provides excellent interactivity and user-friendliness in HSI–MSI fusion applications. The experimental results validate that the framework achieves promising speedups while maintaining high accuracy of HSI–MSI fusion for large-scale remote sensing data. In the future, we will add more effective components and further enhance the capabilities and functionality of our framework.

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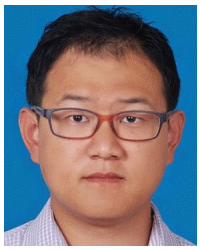
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